

Individual Task 1

This individual assignment combines all topics covered up to and including Week 9. Complete your analysis in the ***Submission template.RMD*** file and submit this file onto Moodle.

Note the following:

- This is a ‘Guided Gen AI Use’ assignment where you are permitted to use generative AI (GenAI) as a supporting tool. Following University guidelines, if GenAI is used, you must submit a short declaration describing the tools used, purpose of use, extent of use, and outline the steps taken to verify the accuracy of the GenAI output.
- You cannot manually modify the data in software like Excel. Anything related to data preparation and analysis must be completed using R.

Data:

The data for this assignment were extracted from Fonterra’s social media accounts (Twitter/X, Facebook, Instagram, YouTube, TikTok, and LinkedIn) in July 2026 using web scraping agents and APIs. It consists of social media content posted or shared by the organisation and associated engagement metrics, such as the number of replies or comments, likes, and shares.

fonterra.csv includes the data to use for this assignment. Refer to later pages for variable names and definitions.

Scenario:

You work as an entry-level analyst in Fonterra’s digital marketing team. Your line manager wants to evaluate the effectiveness of the organisation’s social media strategy across its various social media platforms. You have been asked to analyse content posted or shared on Fonterra’s official social media accounts to identify the types of content that generate the highest levels of engagement.

Your line manager has requested an overview of the topics content creators have been posting about, along with preliminary insights into the factors that may influence user engagement.

Using APIs, you have extracted data on content posted or shared across all of Fonterra’s official social media accounts. You are to complete your analysis using R and report your findings.

Requirement:

1. Maintain a tidy and readable R Markdown file (RMD) that runs without any errors. Annotate your code by including brief descriptions of what the code does. (5 marks)

Examples of annotations: # Import and load packages or # Check and handle missing data

2. Prepare the data for analysis.

- (a) Prepare the non-text data. Briefly note any judgements applied. (15 marks)

Example of a judgement: Removed identical content created at the same time, as these are errors that could skew results

- (b) Create new variables that could be useful in subsequent analyses. Briefly explain why the new variables could be useful. (10 marks)
- (c) Prepare the text data for term frequency analysis and topic modelling. (10 marks)

3. Conduct appropriate and comprehensive analyses of the following areas of interest and provide data-informed insights and recommendations.

Note: Apply basic formatting to any visuals created to make them presentable to a professional audience.

- (a) What topics have content creators been posting about on Fonterra's social media accounts?

Note: 'Topics' refers to the subject matter of the content and is used synonymously with 'themes'.

- i. Term frequency analysis. (10 marks)
- ii. Topic modelling. (15 marks)

Notes: (a) Can try fitting a $k = 6$ model. Fitting a single model should take fewer than 2 minutes. It could take over 30 minutes to run the 'Determining number of topics (k)' lab demonstration code. (b) You may need to undertake additional data preparation if the word-topic probability output is difficult to interpret. Alternatively, try changing the number of k and refit the model. (c) The most probable topic per document can be identified using a pattern such as: `lda_model |> tidy(matrix = "gamma") |> group_by(document) |> slice_max(gamma, n = 1)` and then `left_join()` with the main dataset.

- iii. Commentary on insights from the data analysis. (5 marks)

- (b) What is the level of user engagement with Fonterra's social media content? What factors may influence user engagement?

- i. Exploratory data analysis. (10 marks)

Note: This includes descriptive statistics and visualisations.

- ii. Bivariate analysis. (15 marks)

Note: This includes statistical analyses that assess the relationship between two variables. Limit your analysis to techniques covered in this course (i.e., correlation, t-tests, ANOVA, chi-squared tests, and their non-parametric equivalents).

- iii. Commentary on insights and recommendations from the data analysis. (5 marks)

Variable names and definitions:

Dataset: *fonterra.csv*

Variable name	Description
platform	Social media platform name
url	Content URL
createdAt	Timestamp of when the content was created, based on UTC time; timestamps have 00:00:00 for HH:mm:ss (hour : minute : second) if only yyyy/MM/dd (year / month / day) was available
text	Text included in the content created, extracted from: • Posts for Twitter/X, Facebook and LinkedIn • Caption of the image or video for Instagram and TikTok • Auto-generated or owner-uploaded video caption for YouTube
media	Non-text media included with the content created, could include an image/photo, a video, or a sidecar (i.e., carousel album that contains multiple images and/or videos)
commentCount	Total number of comments (non-Twitter/X) or replies (Twitter/X) left on the content created
shareCount	Total number of times users shared the content created
likeCount	Total number of times users liked (non-TikTok) or digged (TikTok) the content created
videoplayCount	Total number of times users played the video media